POLITECNICO MILANO 1863

DATA SCIENCE LAB

A computational framework to study online conspiracy theories on Reddit after Epstein's death

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Academic year: 2024-2025

We know *what* algorithms do, but not *how* they reshape communities

Conspiracy communities reach millions through platform algorithms.

Who joins?
How long do they stay?
How do they adapt?

→

- **Toxicity** analysis
- **Retention** analysis
- **Linguistic** analysis

Releted work

- Pathways to conspiracy
- Linguistic adaptation
- Intervention effects

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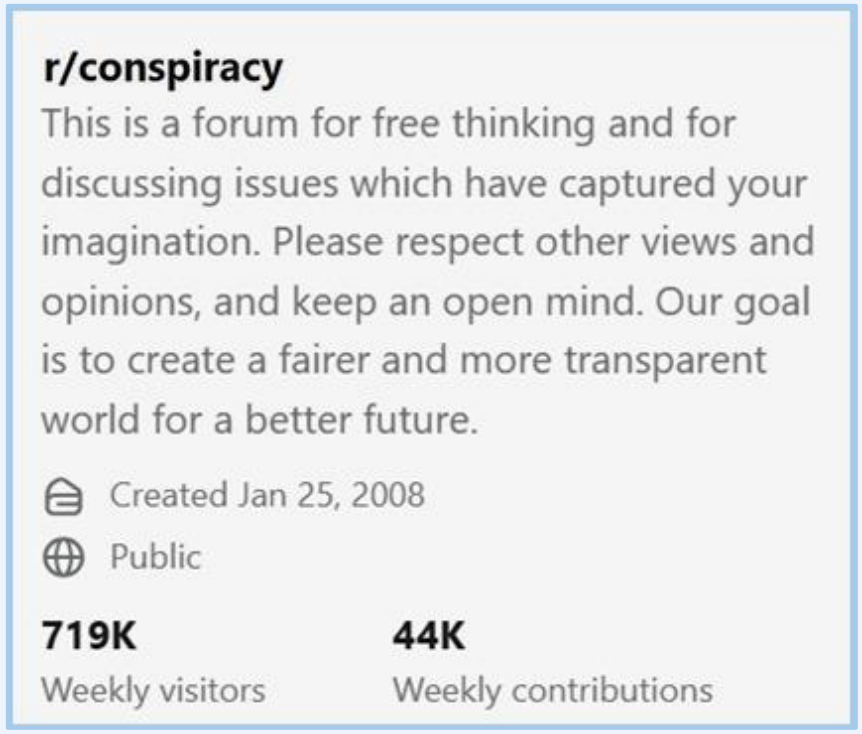
→ Visibility-driven expousure

Introduction

Reddit is a social platform where users share content in **topic-specific communities** called subreddits.

When a post reaches Reddit's homepage, also the users who are not subscribed to that subreddit can see it. **People encounter content they never actively searched for.**



Epstein case

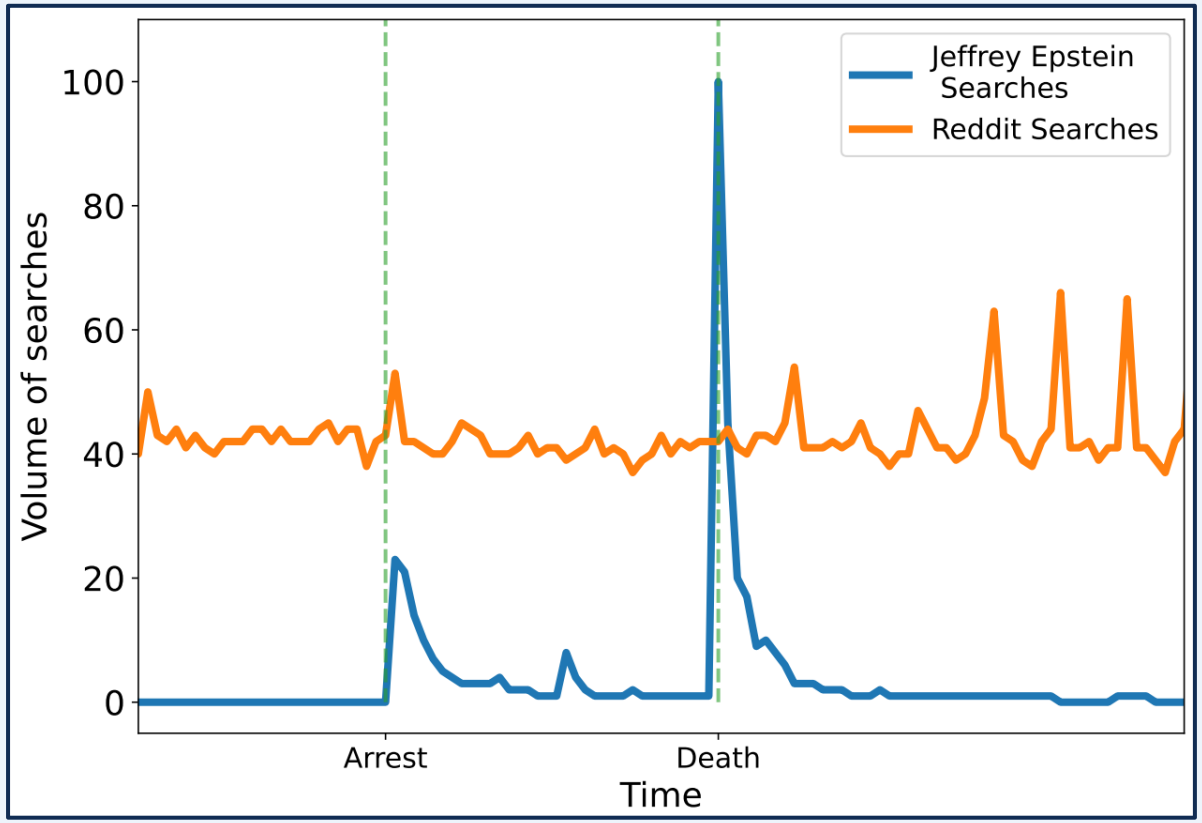
On **August 10, 2019**, Jeffrey Epstein died under **disputed circumstances**.

↓

Fringe conspiracy beliefs aligned with **mainstream** skepticism

↓

r/conspiracy reached Reddit's **homepage**



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r/conspiracy reached Reddit's **homepage**

- Clear before/after boundary
- External event
- No interfering factors

Research questions

RQ1 How do sudden visibility events alter behavioral/linguistic signatures of core members?

RQ2 How does homepage visibility affect community composition and retention?

RQ3 Does timing/context of entry influence linguistic integration?

Data collection

73K comments

2.2M submissions

- All 2019 *r/conspiracy* posts
- Pushshift API
- Filtered out deleted posts

User cohorts

Temporal dimension

Topical dimension	Arrest period – Organic discovery	Death period – Homepage visibility
First post in Epstein-related thread	**Arrest – engaged** (32K)	**Death – engaged** (33K)
First post in other conspiracy topic	**Arrest – not engaged** (103K)	**Death – not engaged** (65K)

Methods

Core user behavior analysis

Toxicity trend + Lexical shift

Core user behavior analysis

Toxicity trend

+

Lexical shift

- Detoxify
- Interrupted Time Series (ITS) regression

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Toxicity trend

+

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- Empath
- Within-user shifts and between-group comparison
- Benjamini-Hochberg

User retention

Kaplan-Meier estimator

- Compute the probability of user remaining active over time
- Suited for right-censored survival times

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→ Engagement trajectories per user cohort

Semantic alignment

Sentence-BERT (SBERT) embeddings

1. Embed all core user posts
2. Compute weekly centroid
3. Compute cosine distance

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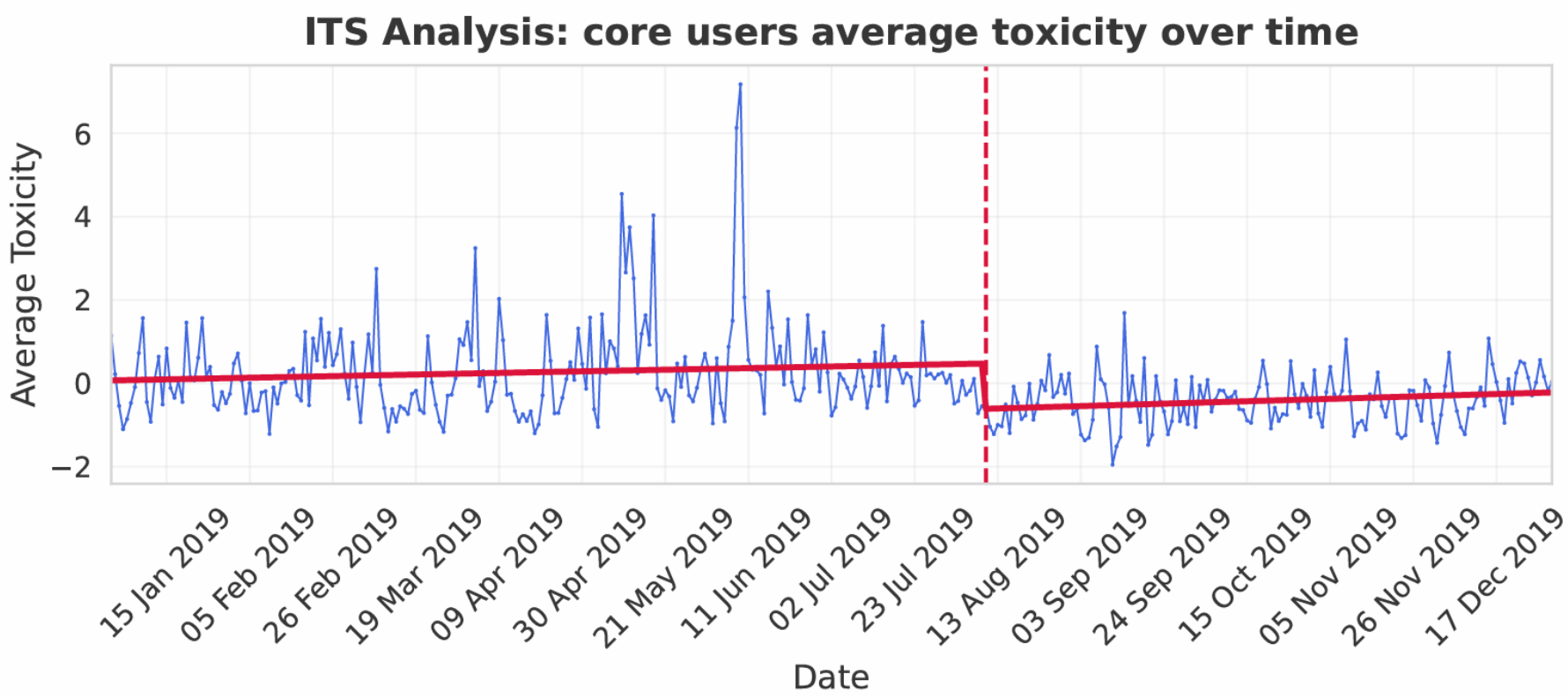
3. Compute cosine distance

→ Weekly semantic distance for each cohort

Findings

RQ1 – Findings



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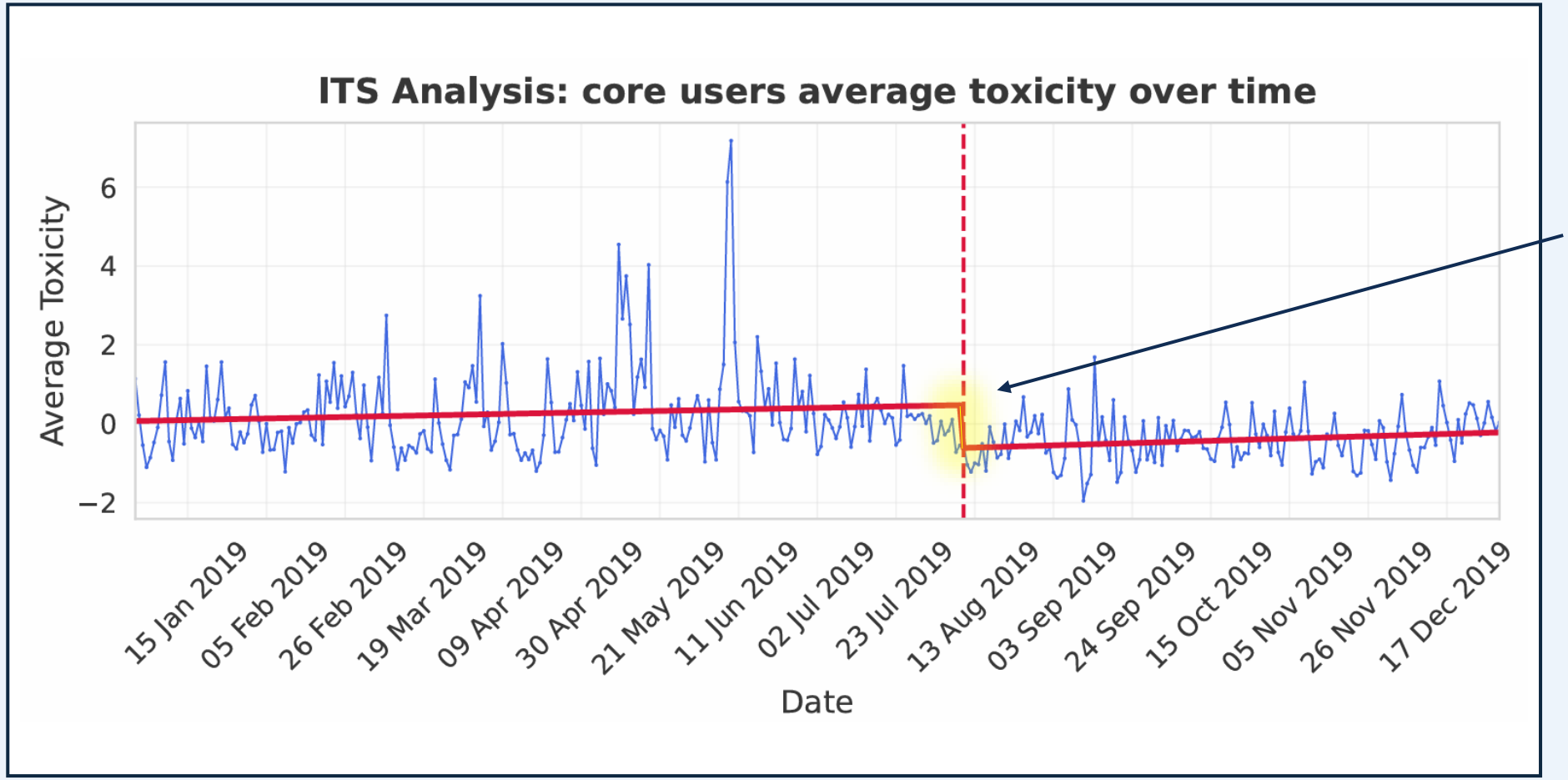
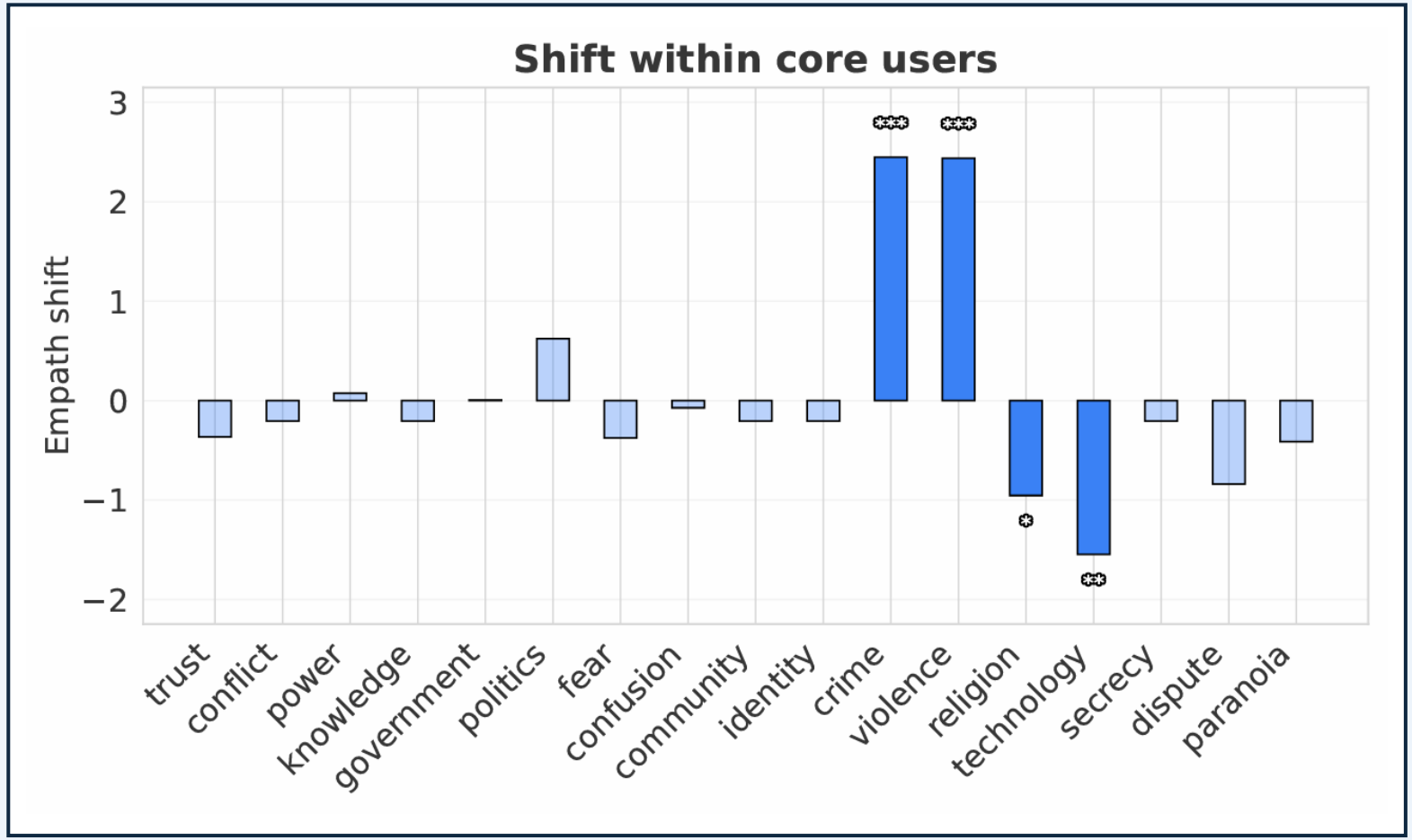


Toxicity significantly dropped ($p < 0.001$) immediately after death

RQ1 – Findings



***: p-val < 0.001
**: p-val < 0.01
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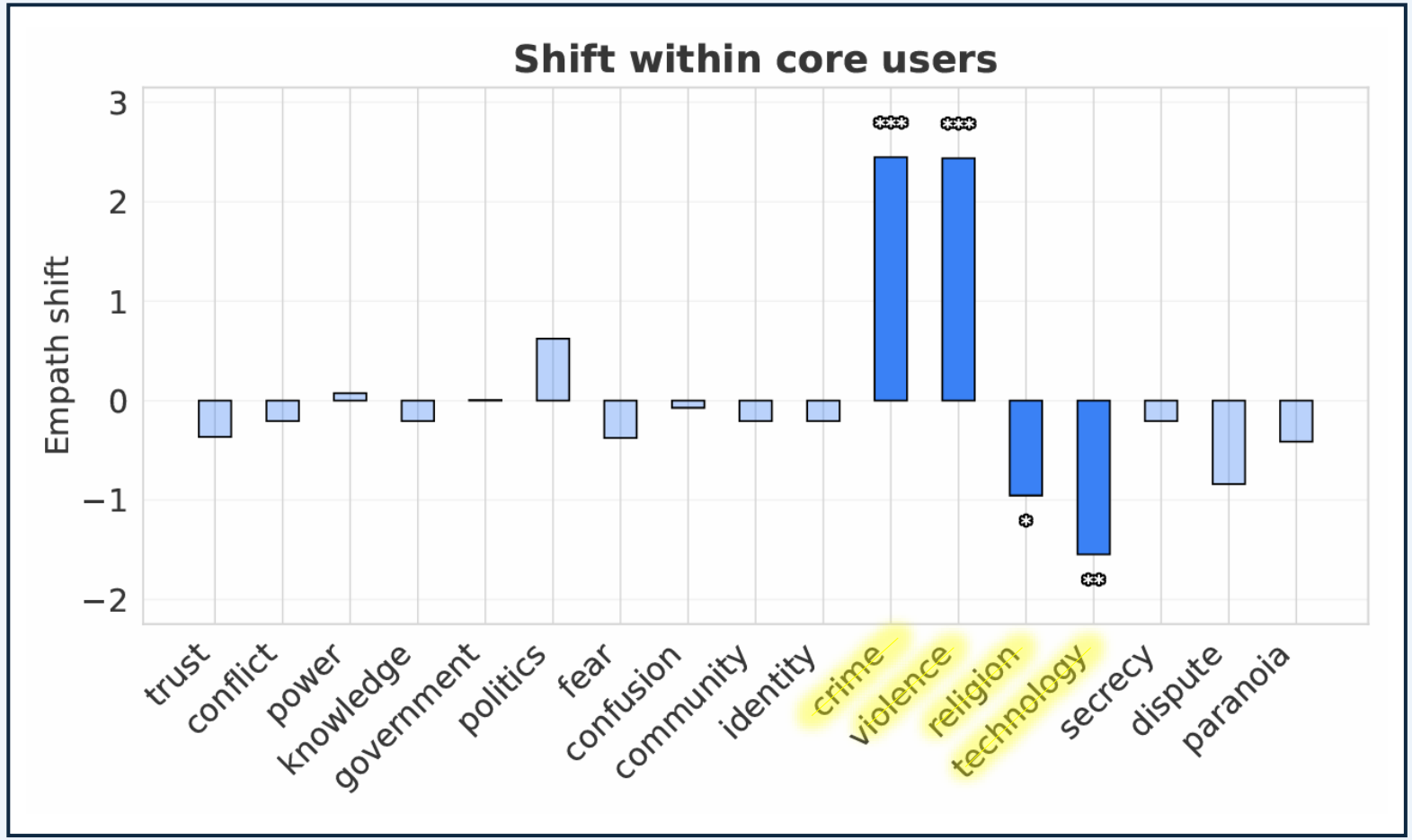

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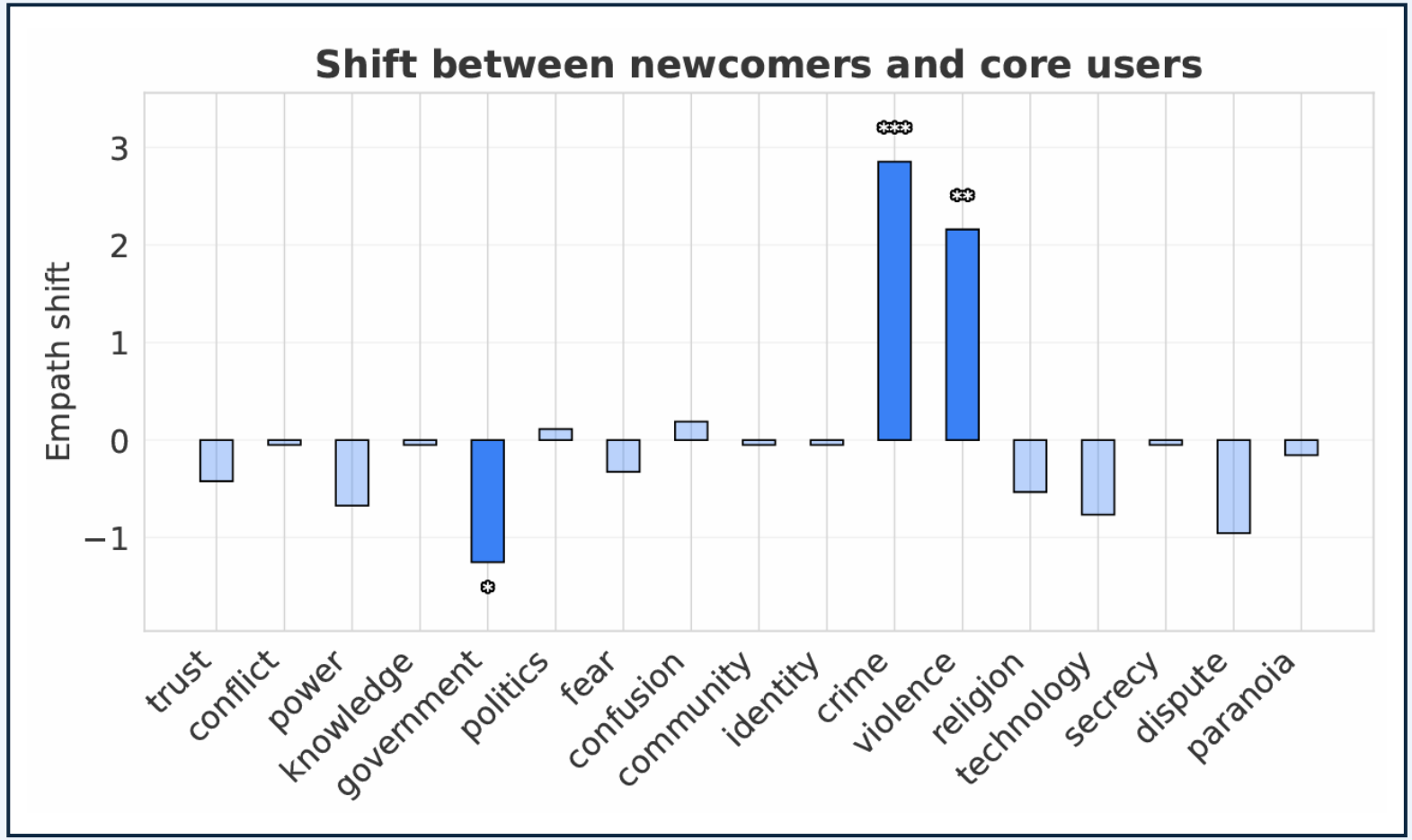
Core users reframed discourse toward criminality and conflict

RQ1 – Findings



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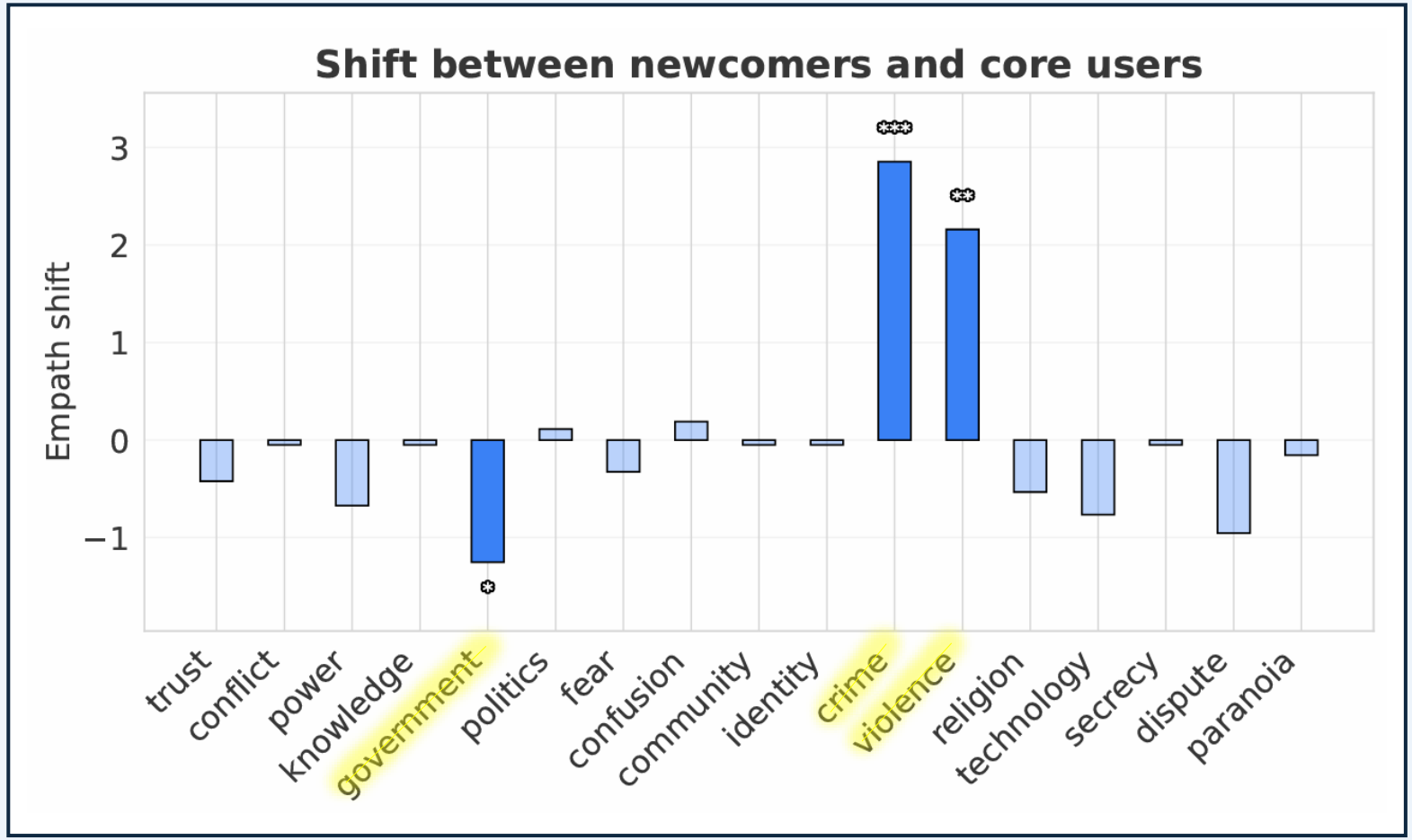

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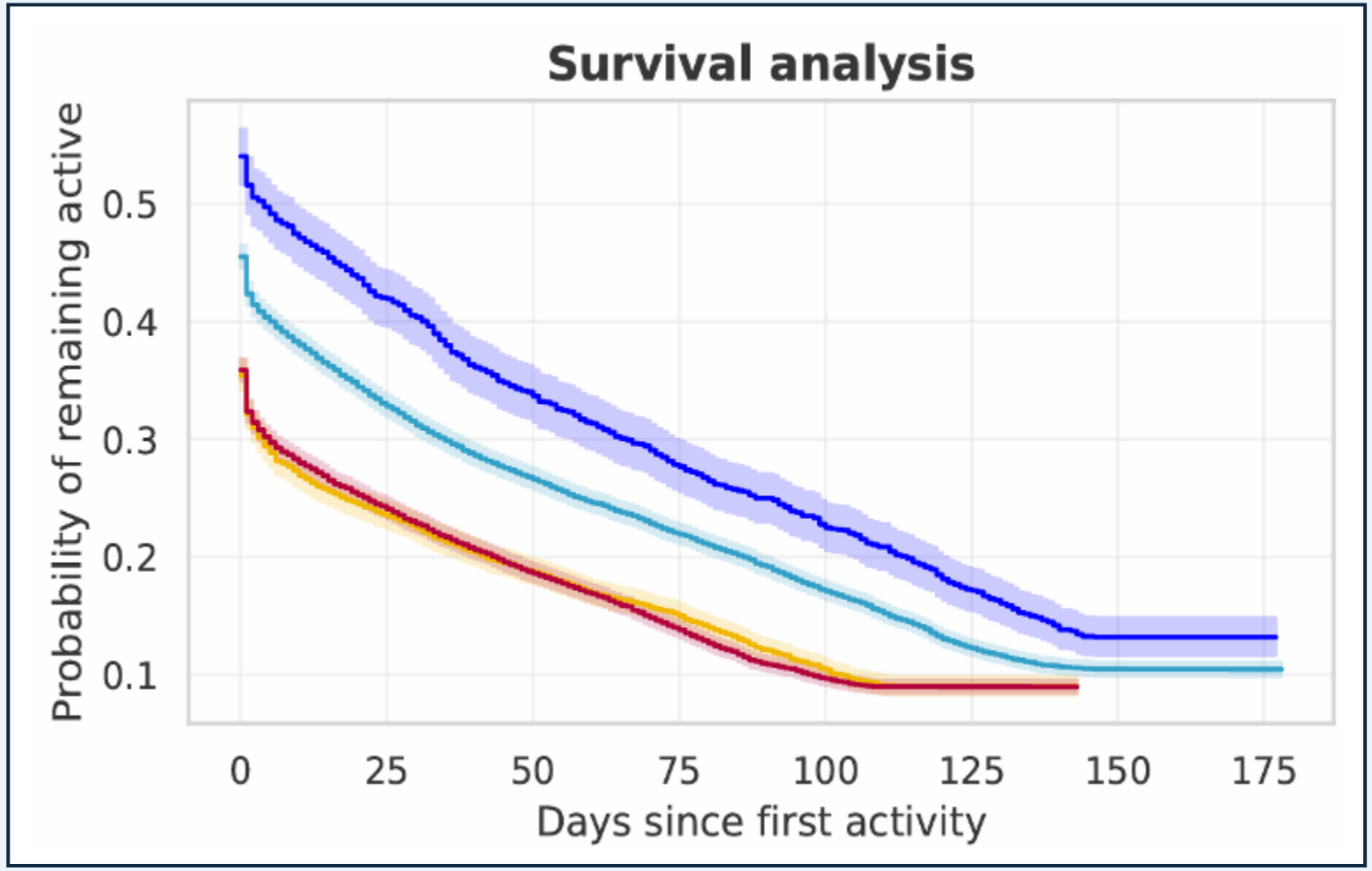
Newcomers introduced sensationalist framing

RQ1 – Findings

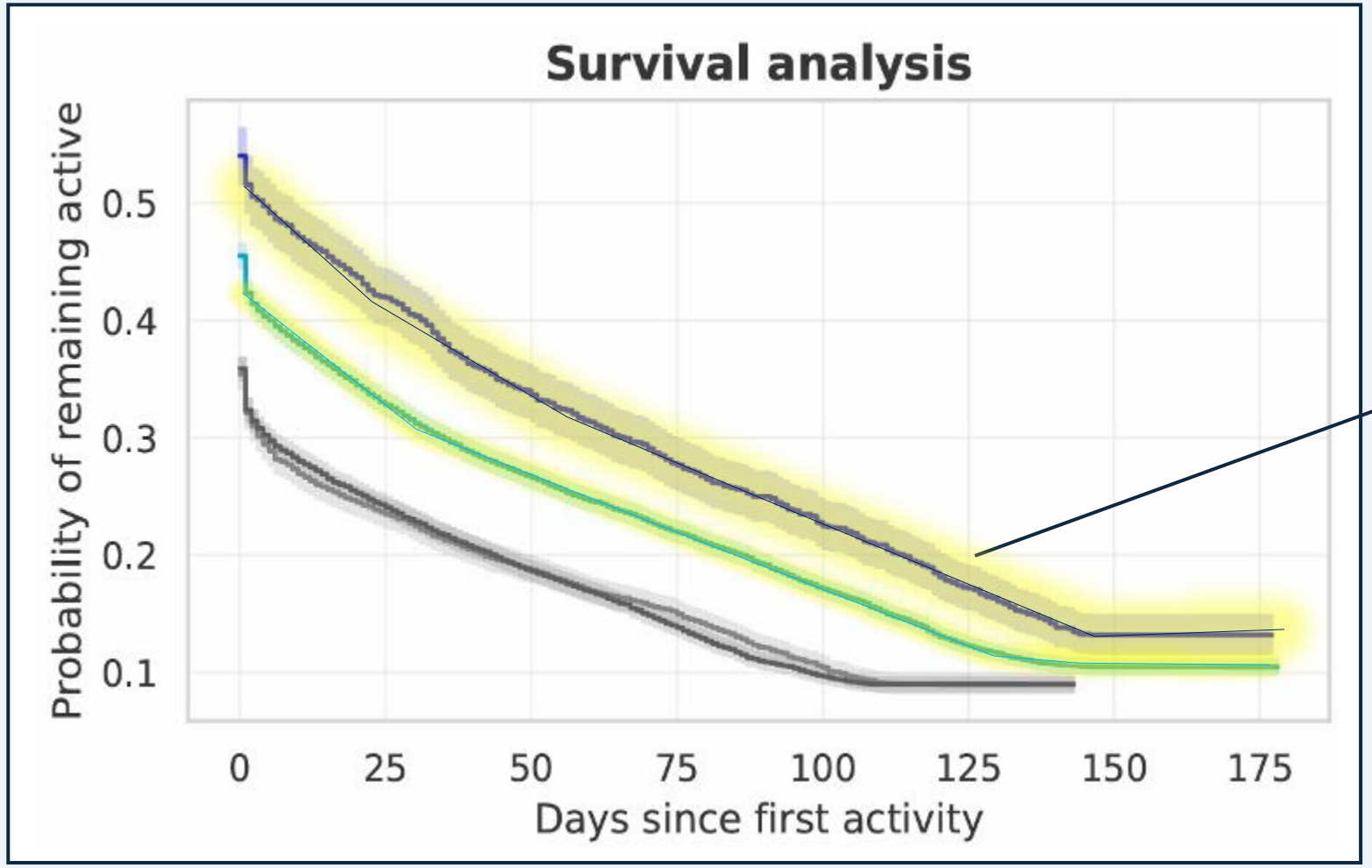
- Core users self-moderated when exposed to wider audience.
- The community evolved, but newcomers did not assimilate.

RQ2 – Findings



Arrest - engaged
Arrest - not engaged
Death - engaged
Death - not engaged

RQ2 – Findings



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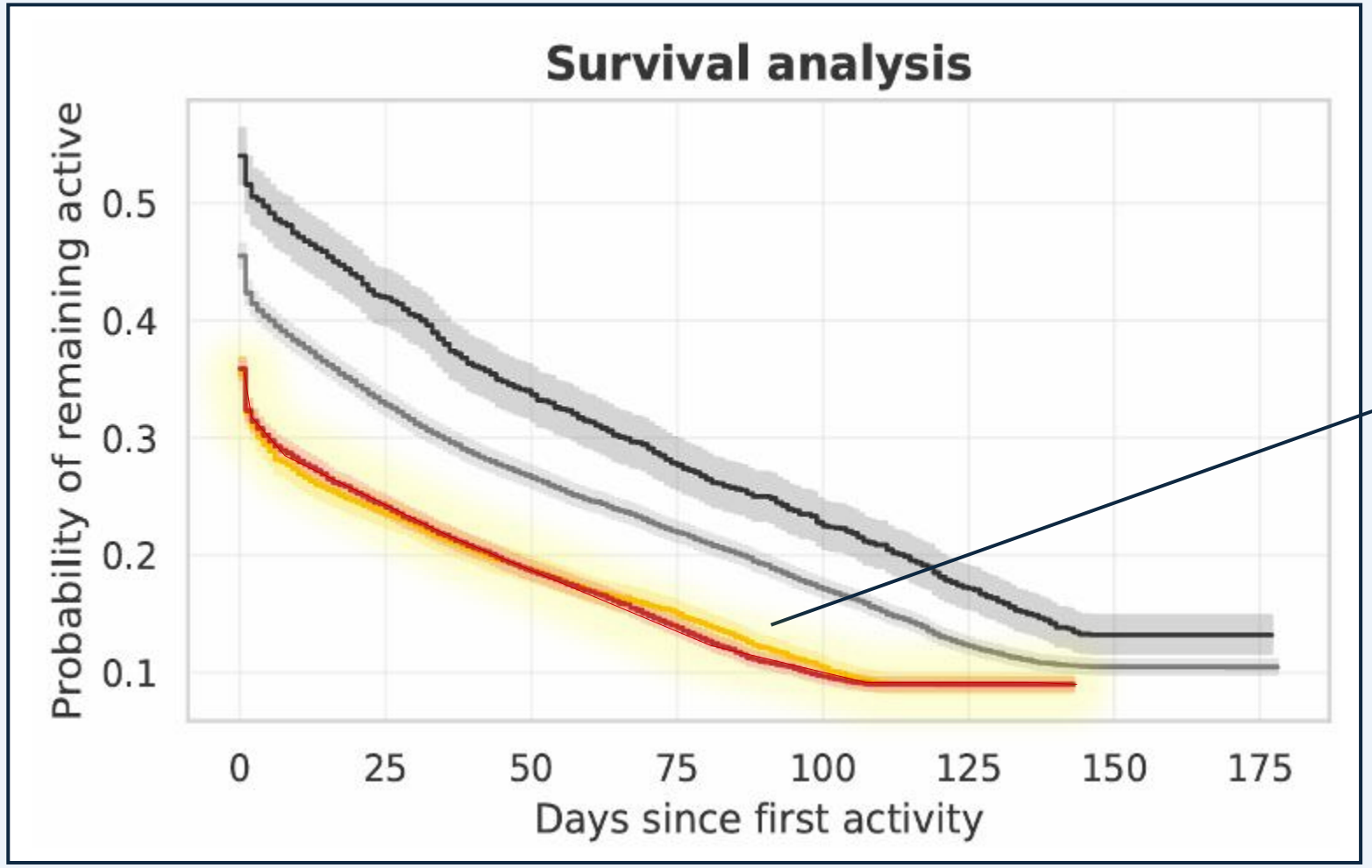
Post arrest cohorts show higher retention

RQ2 – Findings



Arrest - engaged
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Death - engaged
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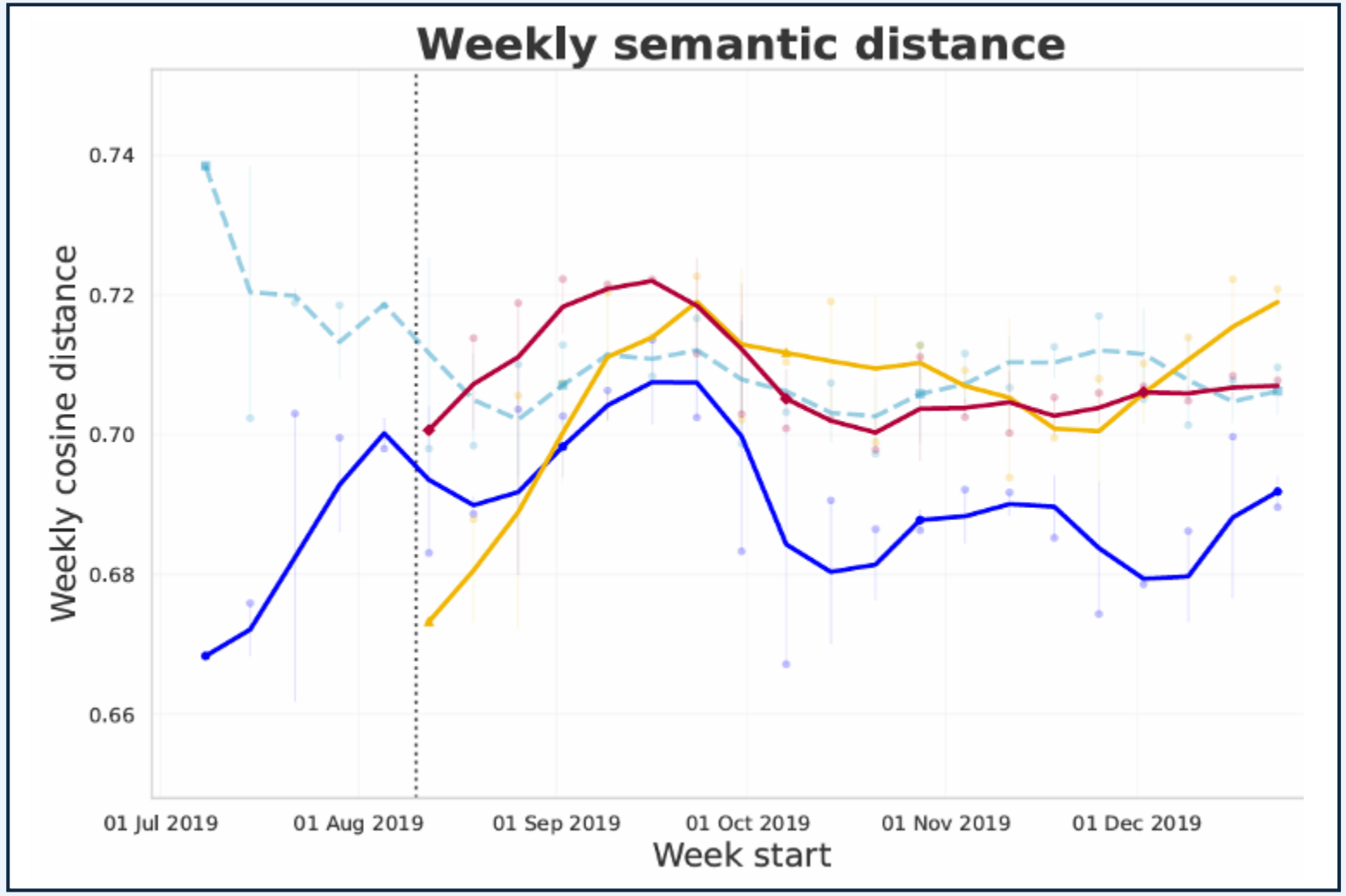
Post death cohorts show faster churn

RQ2 – Findings

- *Organic discovery* → committed, long-term participants.
- *Homepage visibility* → short-lived users who churn rapidly once the attention spike fades.

RQ3 – Findings



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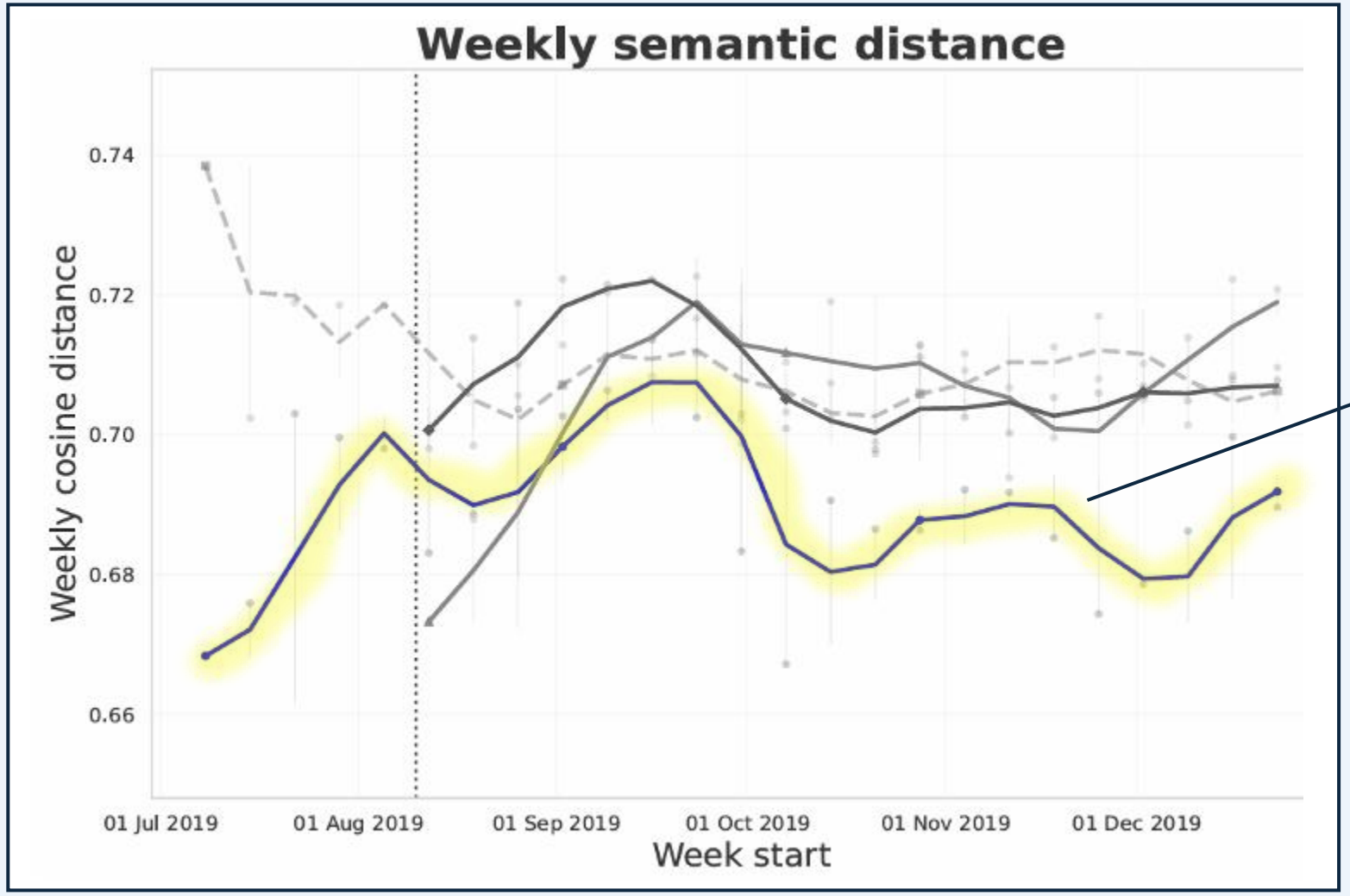

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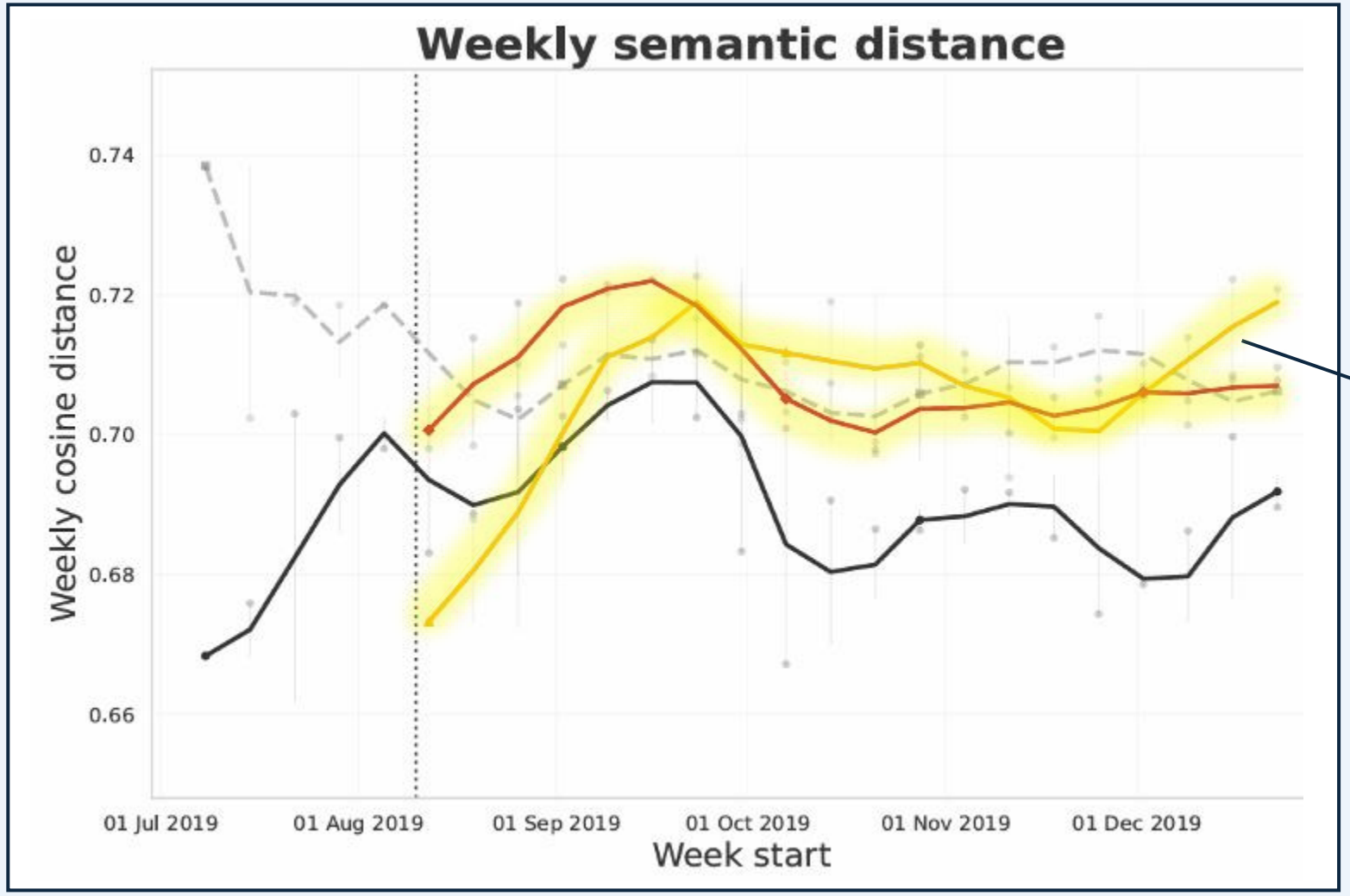
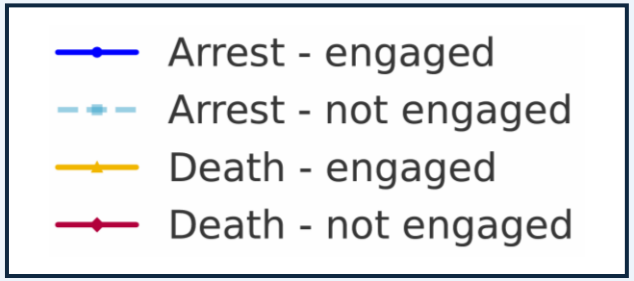
Post arrest cohort is closest to core users' discourse

RQ3 – Findings





Post death cohorts are persistently distant from core users' discourse

RQ3 – Findings

- *Organic discovery* → convergence with core community language.
- *Homepage visibility* → persistent semantic distance; weak linguistic integration.

Conclusions

- Algorithmic visibility **expands the audience count** but **does not organically grow the community** structure.
- **No evidence of durable radicalization** from incidental exposure.

Limitations

- Observation window
- Inferred discovery pathways
- Single community
- Posting activity only

Effects of Algorithmic Visibility on Conspiracy Communities: Reddit after Epstein's 'Suicide'

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Thank you!

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